

Kaiqiang Zheng . Jarvis Consulting

I am a Software and AI Data Engineer with a Master's in Computer Engineering from the University of Toronto and a Bachelor's in Automation Engineering from McMaster University, specializing in AI, software development, and cloud computing. As an AI Web Application Research Associate at UofT, I developed responsive React web applications with Flask backends for real-time AI visualization, built APIs to fetch model outputs, implemented multi-agent LLM pipelines using Python and LangChain, and designed reliable ETL/ELT data pipelines and SQL-based data models on cloud platforms such as Azure to support scalable AI workflows. I am passionate about building scalable, data-driven systems and delivering efficient, production-ready data and AI solutions.

Skills

Proficient: Python, AI/NLP (LLMs, Multi-agent Architecture, OpenAI API, Prompt Engineering, LangChain), Machine/Deep Learning, Linux/Bash, RDBMS/SQL, HTML/CSS, Spark, Databricks, Git, Agile/Scrum

Competent: Java, C++, JavaScript, Docker, Cloud Computing (AWS, Azure, GCP), REST APIs, Flask, Airflow, Maven, JUnit, LaTeX

Familiar: Django, Spring Boot, React, Angular, Hadoop, Postman, Node.js, MATLAB, Figma

Jarvis Projects

Project source code: https://github.com/jarviscanada/jarvis_data_eng_KaiqiangZheng

Cluster Monitor [GitHub]: Developed a Linux cluster monitoring system for Jarvis' LCA team using Docker-contained PostgreSQL to store node performance data. Implemented Bash scripts to automate container lifecycle, database table creation, and data insertion from Linux hardware metrics. Scheduled one-minute interval data collection via crontab and conducted manual testing on GCP virtual machines running Rocky Linux 9. Leveraged Git for version control, centralizing monitoring and reducing manual overhead, enabling administrators to efficiently track system performance and resource usage across multiple clusters.

Core Java Apps [GitHub]:

- **JDBC App:** Developed a stock quote management application that allows users to track their investments, buy, and sell stocks. The application integrates a real-time stock data REST API and stores stock information in a PostgreSQL database via JDBC. A command-line interface (CLI) was created for users to simulate transactions and monitor the performance of their portfolio. Additionally, SLF4J logging was implemented to capture and log any errors that occur during application execution. The project leveraged Maven for efficient dependency management, Docker for containerization and deployment, and was thoroughly tested with JUnit and Mockito to ensure reliability and high code quality throughout the development process.
- **Grep App:** Developed a Java-based application leveraging object-oriented programming principles, designed to function similarly to the Linux grep command. The tool recursively traverses user-specified directories, identifies matches based on user-provided regex patterns, and stores the results in an output file. Java Streams and Lambda expressions were utilized to enhance performance, while Logback was employed for efficient logging. The application was manually tested and debugged using IntelliJ IDEA to ensure reliability. For deployment, a Docker image was created via a custom Dockerfile, with Maven handling dependency management and Git used for version control.

Python Data Analytics [GitHub]: Delivered data-driven insights for London Gift Shop (LGS), a London-based online retailer, by transforming historical sales data into actionable recommendations. Built a PostgreSQL data warehouse and performed analysis in Python (NumPy, Pandas, Matplotlib) via Jupyter Notebook. Key tasks included tracking monthly sales and customer growth, and applying RFM (Recency, Frequency, Monetary) segmentation to support targeted marketing. Ensured reproducibility through Docker containerization and maintained workflow with Git version control.

Spark [GitHub]: Built a distributed data analytics pipeline using Apache Spark (PySpark) to efficiently process large customer transaction datasets for a marketing team. Assessed and compared Zeppelin (Hadoop-based) and Databricks (AWS-based) Spark environments to evaluate scalability and processing efficiency. Utilized Delta Lake on Databricks and HDFS on Zeppelin, with Hive Metastore for metadata and schema management. Conducted transaction and retention analysis using Spark's structured APIs, delivering insights that improved marketing segmentation and targeting strategies.

Equity Market ETL Pipeline [GitHub]: Delivered an end-to-end batch ETL pipeline for U.S. equity and ETF market data using Apache Airflow and Apache Spark (PySpark), transforming raw ticker-level OHLCV files into analytics-ready datasets. Implemented a medallion architecture (Bronze / Silver / Gold) with schema enforcement, data quality validation,

and enrichment using company metadata. Orchestrated ingestion, transformation, and validation workflows with Airflow, and optimized scalability through partitioned Parquet storage. Designed the pipeline to be CI/CD-ready with unit tests and containerized execution using Docker, enabling reproducible and production-aligned data processing.

Highlighted Projects

SherlockGPT: Root Cause Analysis with Large Language Model [GitHub]: Developed an automated Root Cause Analysis (RCA) engine leveraging large language models (GPT-4 via OpenAI API) and prompt engineering to identify incident root causes from cloud system logs and metrics. Designed a multi-agent architecture that iteratively analyzes and ranks potential causes. Utilized the LEMMA-RCA dataset for evaluation, enabling improved incident triage and decision-making efficiency.

Sentiment Analysis for Customer Service Ticket Analysis [GitHub]: Built and deployed a sentiment analysis application in Docker to classify customer service ticket sentiments. Integrated OpenAI API for text analysis and Gradio for an interactive web interface, enabling real-time visualization of sentiment trends and improving customer feedback insights.

Full Stack Semester Project: Designed and implemented a full-stack web application using Django, Python, and MySQL for backend logic and data management. Built a responsive frontend using HTML, CSS, Bootstrap, and JavaScript. Containerized the application with Docker for consistent deployment, demonstrating proficiency in full-stack development and cloud deployment workflows.

Professional Experiences

Data Engineer, Jarvis (2025-present): Developed and maintained scalable software solutions within an Agile/Scrum framework using Linux, Bash, SQL, Git, Java, and Docker. Contributed to multiple projects, designing and implementing efficient backend systems, optimizing database performance, and streamlining deployment processes. Participated in sprint planning, daily stand-ups, and retrospectives to ensure iterative progress and high-quality delivery. Leveraged big data technologies such as Apache Spark and Databricks to handle large datasets and build robust data processing pipelines.

AI Web Application Research Associate, University of Toronto SG (2024): Developed and maintained responsive web applications using React, HTML, CSS, and integrated Python Flask backends to visualize AI model outputs in real time. Collaborated with graduate researchers to design intuitive UI/UX interfaces and integrate SQL databases for experimental data storage. Implemented multi-agent NLP pipelines using Python and OpenAI API, supported scalable AI workflows, and followed Git-based version control and regular code reviews to ensure robust software practices.

Design Technician, University of Toronto SG (2022-2023): Utilized software tools to code programs and perform calculations, creating accurate design drawings for engineers and ensuring compliance with specifications. Supported busbar design through programming to assist assembly processes. Managed administrative tasks including enrolling new employees and maintaining records, contributing to smooth operational workflows.

Education

University of Toronto (2024-2025), Master of Engineering, Electrical and Computer Engineering - GPA: 3.83/4.0

McMaster University (2019-2023), Bachelor of Engineering Technology, Automation Engineering - Graduated with Distinction - Scholarship for Academic Excellence - Dean's List (2021, 2022) - GPA: 3.70/4.0

Miscellaneous

- AWS Academy Cloud Foundations Badge (2023)
- FANUC CERT Handling Tool Operations and Programming (2022)
- 3rd Place for Best Design in the International Design and Engineering Education Association (IDEEA) Competition (2023)
- Teaching Assistant: Provided one-on-one tutoring in mathematics to summer students, covering algebra and calculus.
- Fall ECE Graduate Orientation: Assisted the ECE Graduate Office by participating in a mix-and-mingle reception, addressing student inquiries, and providing information and guidance about the program.
- Talent Show: Completed a wonderful performance by playing the guitar while maintaining a hard study state.